Graduate Graph Theory 2026
Final Assignment

- Do **five** of the following problems. Only one of the five can be a Bonus problem. (25 marks)
- Solutions, not just answers. Be clear and concise. I am marking these both on the ideas and on how you write them. If you know your handwriting is hard to read, please type your solutions.
- Due in by midnight on June 17. By hand or e-mail.

Problems

1. How many edges must you remove from the Petersen graph to make it planar?
2. The crossing number of a graph is the minimal number of edge crossings in a drawing of the graph in the plane. Show that the crossing number can be greater than the minimum number of edges you must remove from the graph to make it planar.
3. Find a graph G such that $\chi(G) = 2$ and $\chi_\ell(G) \geq 4$.
4. Show that an interval graph is a proper interval graph if and only if it does not contain an induced subgraph isomorphic to $K_{1,3}$.
5. A vertex v in a chordal graph is G is a *simplicial keystone* if there is a simplicial ordering of G in which v is the last vertex. What is that maximum d such that any large enough chordal graph has d simplicial keystones.
6. Show that if $p = o(\frac{\log n}{n})$ then a.a.s. $G_{n,p}$ has isolated vertices.

Bonus

7. Make up, and solve, a problem that is non-trivial and appropriate to the notes. (If you find the problem somewhere, reference it.)
8. Find a mathematical mistake in the notes, and fix it. There are plenty, but if you can't find one, find a part of the notes that is unclear, and rewrite it to make it clear.